

Foundation (Jalapeno)

Q1. What is the solution to the inequality $|x - 3| < 2$?

- (A) $1 < x < 5$
 (B) $-1 < x < 5$
 (C) $0 < x < 4$
 (D) $2 < x < 6$
 (E) $-2 < x < 4$

Q2. The distance between two planets is $|x + 5| + 2$ miles. If the distance is less than 11 miles, what is the range of x ?

- (A) $-8 < x < 4$
 (B) $-6 < x < 2$
 (C) $-4 < x < 6$
 (D) $-3 < x < 5$
 (E) $-2 < x < 8$

Q3. The absolute value of the force F applied to an object is given by $|F - 10| > 5$. What are the possible values of F ?

- (A) $F < 5$ or $F > 15$
 (B) $F < 10$ or $F > 20$
 (C) $F < 0$ or $F > 10$
 (D) $F < 15$ or $F > 5$
 (E) $F < 20$ or $F > 0$

Q4. The inequality $|x| > 4$ can be written in interval notation as

- (A) $(-\infty, -4) \cup (4, \infty)$
 (B) $(-\infty, 4) \cup (4, \infty)$
 (C) $(-4, 0) \cup (0, 4)$
 (D) $(-4, 4)$
 (E) $(-\infty, 0) \cup (0, \infty)$

Q5. A spaceship is traveling at a speed of $|v - 20|$ km/s. If the speed is greater than 10 km/s, what are the possible values

of v ?

- (A) $v < 10$ or $v > 30$
 (B) $v < 20$ or $v > 10$
 (C) $v < 30$ or $v > 10$
 (D) $v < 10$ or $v > 40$
 (E) $v < 40$ or $v > 0$

Q6. The inequality $|x + 2| \leq 3$ can be solved by

- (A) graphing the function $y = |x + 2|$ and finding the x -values where $y \leq 3$
 (B) finding the x -values where $-3 \leq x + 2 \leq 3$
 (C) solving the equation $x + 2 = 3$
 (D) graphing the function $y = x + 2$ and finding the x -values where $y \leq 3$
 (E) finding the x -values where $x + 2 = 0$

Q7. The temperature T of a planet is given by $|T - 20| < 5$. What are the possible values of T ?

- (A) $15 < T < 25$
 (B) $10 < T < 30$
 (C) $20 < T < 40$
 (D) $0 < T < 10$
 (E) $-5 < T < 45$

Q8. A lab experiment involves measuring the magnitude of a force F with an error of $|F - 10| > 2$. What are the possible values of F ?

- (A) $F < 8$ or $F > 12$
 (B) $F < 10$ or $F > 20$
 (C) $F < 0$ or $F > 10$
 (D) $F < 12$ or $F > 8$
 (E) $F < 20$ or $F > 0$

Open Ended Questions (FRQ)

Q9. The distance between two asteroids is given by $|d - 100| + 20$ miles. If the distance is less than 150 miles, find the range of d and express your answer in interval notation.

Q10. The speed v of a spacecraft is given by $|v - 50|$ km/s. If the speed is greater than 30 km/s, find the possible values of v and graph the solution set on a number line.

Chef's Tips

- Tip 1: Use a number line to visualize the solution sets of absolute value inequalities.
- Tip 2: Remember that the absolute value of a quantity is always non-negative.
- Tip 3: When solving absolute value inequalities, consider both the positive and negative cases.